NIVEL I

RESOLUÇÃO SEMANA 4 - MAT. 3

1) a) $\Delta ABC \cong \Delta EFG$ (L.A.L)
b) $\Delta ABC \cong \Delta MNP$ (L.L.L)
c) $\Delta EFG \cong \Delta MNP$ (A.L.A)
d) $\Delta ACB \cong \Delta QPR$ (L.A.L)
e) $\Delta PQM \cong \Delta TSR$ (A.L.A)
f) $\Delta LJK \cong \Delta DFA$ (L.A.A)

2) a) $x = \frac{70°}{2} = 35°$

b) $x = \frac{150° + 50°}{2} = 100°$

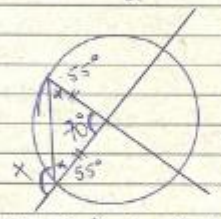
c) $x = \frac{120°}{2} = 60°$

d) $x = \frac{100° - 50°}{2} = 25°$

e) $x = \frac{105° - 65°}{2} = 50°$

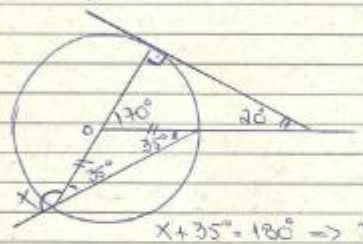
f) $x = \frac{110° - 70°}{2} = 20°$

3) a)



$x + 55° = 180° \Rightarrow x = 125°$

b)



$x + 35° = 180° \Rightarrow x = 145°$

4) a) $2x = 140°$
$x = 70°$

b) $3x = 30°$
$x = 10°$

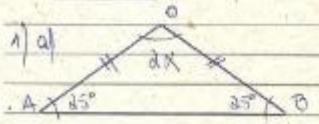
5) a) $x + 110° = 180°$
$x = 70°$

b) $x + 82° = 180°$
$x = 98°$

c) $x + 30° = 180°$
$x = 150°$

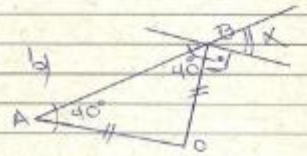
NIVEL II

1) a)



$2x + 50° = 180°$
$x = 65°$

b)



$x + 90° + 40° = 180°$
$x = 50°$

2) a)

$2y + 120° = 180°$

$y = 30°$

$x + y = 90°$

$x = 75°$

b) $\widehat{AB} = \widehat{DE} = \frac{360°}{5} = 72°$

$x = \frac{72° + 72°}{2}$

$x = 72°$

3) $\widehat{PB} = 50°$ $\quad \alpha = \frac{\widehat{PB}}{2} \Rightarrow \alpha = 25°$

4) $\widehat{BE}$ (maior) $= 2\alpha \Rightarrow \widehat{BE}$ (menor) $= 360° - 2\alpha$

$\widehat{CE}$ (maior) $= 2\beta \Rightarrow \widehat{CE}$ (menor) $= 360° - 2\beta$

$\widehat{BE}$ (menor) + $\widehat{CE}$ (menor) + $\widehat{BC}$ (menor) $= 360°$

$360° - 2\alpha + 360° - 2\beta + 70° = 360°$

$\alpha + \beta = 215°$

5) $CE = CF = 7$

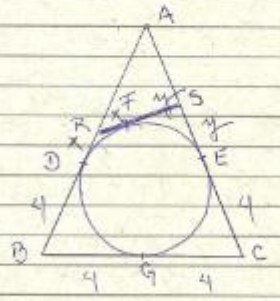
$AF = AD = 5$

$DB = BE = 8$

logo: $2p = 2 \cdot 7 + 2 \cdot 5 + 2 \cdot 8$

$2p = 40$

6)



$BG = CG = EC = DB = 4$

$AD = AE = 20 - 4 = 16$

$AS = 16 - y$

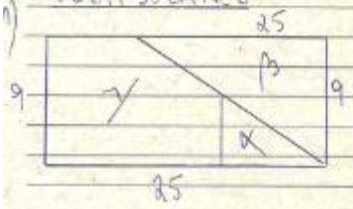
$AR = 16 - x$ $\Rightarrow RS = x + y$

$2p(\Delta ARS) = 16 - x + 16 - y + x + y$

$2p(\Delta ARS) = 32$

VESTIBULARES

7)



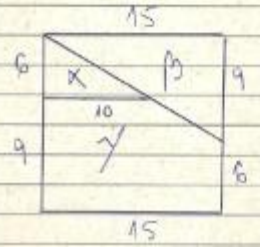
$S = 25 \cdot 9$

$S = 225$

$225 = 15^2$

logo:

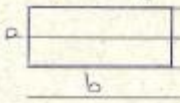
$l = 15$



$S_\alpha = \frac{6 \cdot 10}{2}$

$S_\alpha = 30$

2) retângulo AMARELO



$b + 2y = 15$ ✓

$3a = 15$ $\quad b + 2a + x = 26 \rightarrow b + x = 16$

$a = 5$ $\quad 2b + x + 1 = 26 \rightarrow b + b + x + 1 = 26 \therefore b = 9$

então: $y = 3$ e $x = 7 \Rightarrow x - y = 4$

3)

a) $\operatorname{sen} x = \frac{6400}{12800} = \frac{1}{2} \Rightarrow x = 30°$ então $A\hat{O}B = 120°$

$\begin{cases} 360° \longrightarrow 2\pi R \\ 120° \longrightarrow \overset{\frown}{AB} \end{cases}$ $\quad \overset{\frown}{AB} = \frac{2\pi \cdot 6400}{3}$

$\overset{\frown}{AB} = \frac{12800\pi}{3}$ Km

b)

Teorema dos Cossenos:

$d^2 = 12800^2 + 6400^2 - 2 \cdot 12800 \cdot 6400 \cdot \cos\theta$

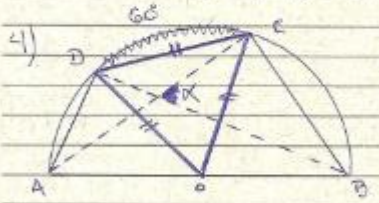
$d^2 = (2x)^2 + x^2 - 2 \cdot 2x \cdot x \cdot \cos\theta$

$d^2 = 4x^2 + x^2 - 4x^2 \cdot \frac{3}{4}$

$d^2 = 2x^2$

$d = x\sqrt{2} \Rightarrow d = 6400\sqrt{2}$ Km

4)



$A\hat{D} + D\hat{C} + C\hat{B} = 180°$

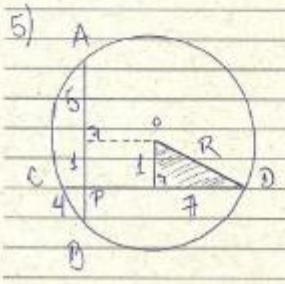
$A\hat{D} + 60° + C\hat{B} = 180°$

$A\hat{D} + C\hat{B} = 120°$

$x = \frac{A\hat{D} + B\hat{C}}{2}$

$x = 60°$

5)



$AP \cdot PB = CP \cdot PD$

$6 \cdot 4 = 2 \cdot PD$

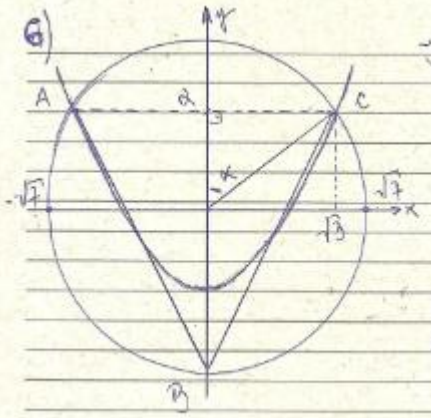
$PD = 12$

$R^2 = 1^2 + 7^2$

$R^2 = 50$

$R = 5\sqrt{2}$

6)



$y + 1 = x^2 \Rightarrow x^2 + y^2 = 7$

$y + 1 + y^2 = 7$

$y^2 + y - 6 = 0 \begin{cases} y = -3 \text{ (N.C)} \\ y = 2 \end{cases}$

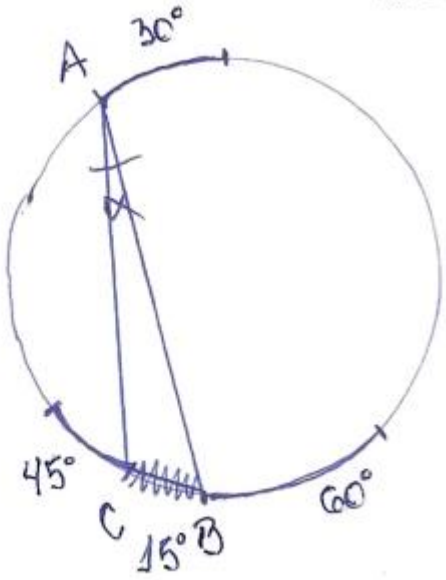
$y = 2 \Rightarrow x = \sqrt{3}$

$\operatorname{tg} x = \frac{\sqrt{3}}{2} = \frac{3}{2\sqrt{3}}$ logo: $x = 41°$

$\overset{\frown}{AC}$ (menor) $= 2x$

$A\hat{B}C = x \Rightarrow A\hat{B}C = 41°$

7)



$\alpha = \frac{15°}{2}$

$\alpha = 7{,}5°$